



Interactive Problems

Problem Solving

- Anikate Koul

Problem 1



Searching Local Minima: <https://codeforces.com/contest/1479/problem/A>

Main Idea

$a[i]$ $a[i+1]$



- In one iteration query for three values $a[i-1]$, $a[i]$ and $a[i+1]$.
- Depending upon the resultant values of these indices, decide in which direction the possible solution can lie in.
- In every iteration, we would be rejecting either the left or the right part of the array (with one end at index i).
- Thus, it acts like binary search.

$$a = [\overset{\downarrow}{\underset{=}{a_1}}, a_2, a_3, \dots, \overset{\downarrow}{\underset{=}{a_n}}]$$

$$\underbrace{a_0 = +\infty}_{\uparrow}, \quad \underbrace{a_{n+1} = +\infty}_{\uparrow}$$

$$l = 1, \quad r = n,$$

$$mid = (l + r) / 2$$

→ ↑

$$(l \leq r),$$

$$(\underbrace{a[mid-1]}_{+\infty}, \underbrace{a[mid]}_{\substack{a_2 \\ n}}, \underbrace{a[mid+1]}_{+\infty})$$

mid → Possible local minimum,

a[mid],

$a[mid-1] > a[mid] < a[mid+1]$ $\rightarrow$ ①
local minimum (mid) ✓

$a[mid-1] < a[mid] < a[mid+1]$ $\rightarrow$ ②,,

$a[mid-2] < a[mid-1]$

$a[mid-3] < a[mid-2]$,,

$a[mid-4]$,, -

$a[1]$ $< a[2] < a[3] \dots < a[mid-2] < a[mid-1]$
 $< a[mid]$

Local minimum,,

$[1, mid-1]$,,

$$a[mid-1] > a[mid] > a[mid+1] \rightarrow (3)$$

$[mid+1, n]$,

$$\underline{a[mid-1] < a[mid] > a[mid+1]} \rightarrow (4)$$

← left. $a[mid-2] > a[mid-1] \leftarrow mid-1 \rightarrow \underline{\underline{ans}}$
 $a[mid-2] < a[mid-1] : \leftarrow$

$$\underline{a[mid-2] < a[mid-1] < a[mid]},$$

$[1, mid-2]$,

$\{6, 5, 1, 2, 4, 3, 7\}$
 1 2 3 4 5 6 7
 ↑

$l=1, r=7$
 (1) $mid = (1+7)/2 = 4$

$[3, 4, 5]$
 ↓
 $[1, 2, 4]$
 val
 $(2, 3)$
 $(1, 3)$

↓ ↓
 T T T F F F

T → left

F → right

① → $mid-1 > \underline{mid} < mid+1$

$mid-1 < mid < mid+1 \rightarrow (T)$

② $mid-1 < mid < mid+1 \rightarrow (F)$

③ $mid-1 > mid > mid+1 \rightarrow (F)$

④ $mid-1 < mid > mid+1 \rightarrow (F)$

Problem 2

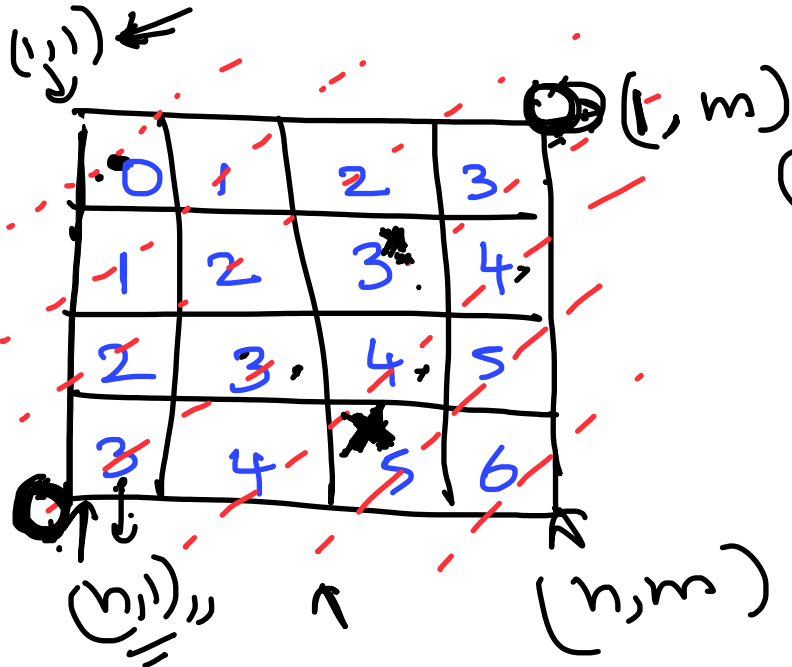


Find A Mine: <https://codeforces.com/contest/1934/problem/C>

Main Idea



- Firstly query from the position $(1, 1)$.
- This gives the idea of the diagonal on which at least one of the mine lies.
- Then query from the two endpoints of the diagonal and at least one of them will point to the mine in that diagonal.
- So, in total we utilize **4** queries.



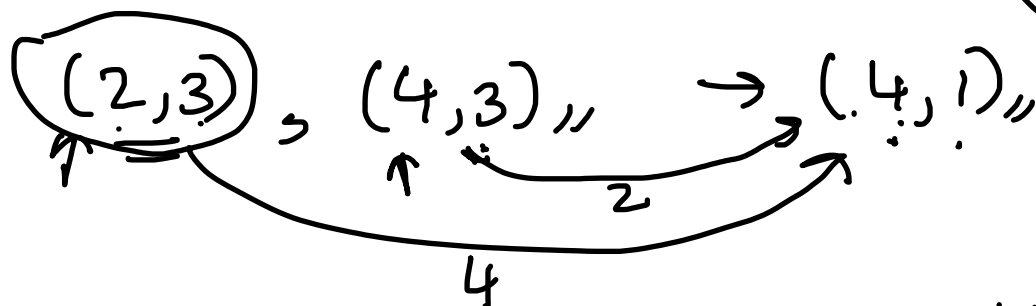
(x_1, y_1)

(x_2, y_2)

$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

→ Manhattan.
 $|x_1 - x_2| + |y_1 - y_2|$

min. manhattan distance



(3,2)

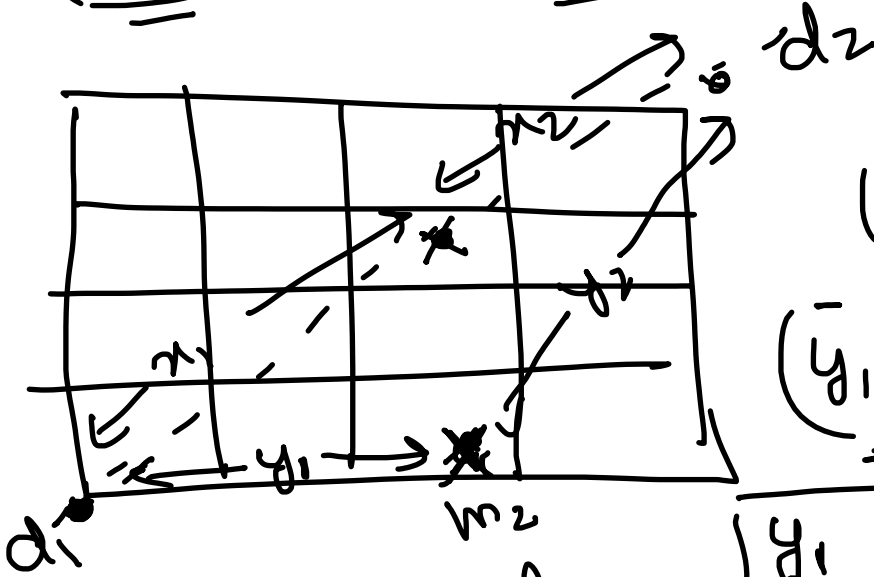
d_1 ←
 d_2 ←
 dist_1
 dist_2
 mine on diagonal
 $d_1 \leftrightarrow d_2$

mine on diagonal → (x_1, d_1)
 (x_2, d_2)

mine not on diagonal → (y_1, d_1)
 (y_2, d_2)

$$(x_1 + x_2) \leq - \quad (y_1 + y_2) \leq -$$

$$(y_1 + y_2) > (x_1 + x_2) \quad \text{manhattan dist} \geq \text{euclidean dist.}$$



$$(y_1), (y_2), (x_1 + x_2)$$

$$(y_1 + y_2) > (x_1 + x_2) \leftarrow$$

$$y_1 < x_1 \leftarrow$$

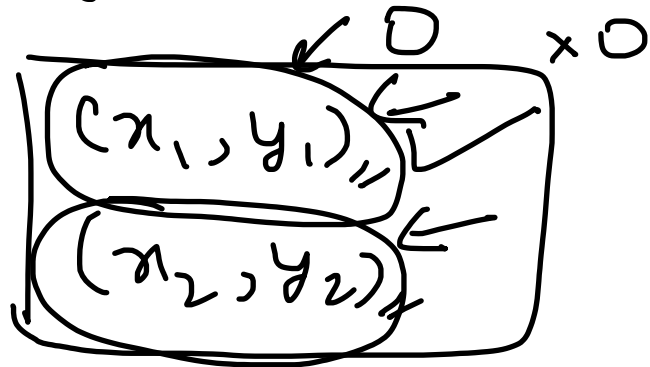
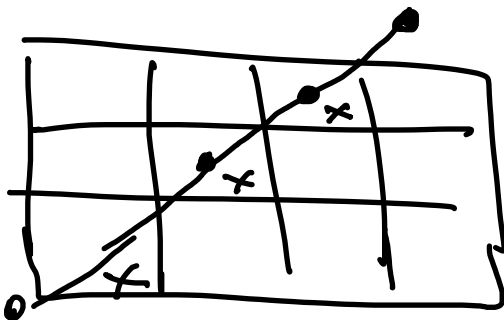
$$x_2 < y_2 \leftarrow$$

✓
(1,1) → diagonal,

✓
(d1) → dist₁
(d2) → dist₂

at least 1.

mine on diagonal



$$(1, 1) \rightsquigarrow a$$

$$x + y = a + 2$$

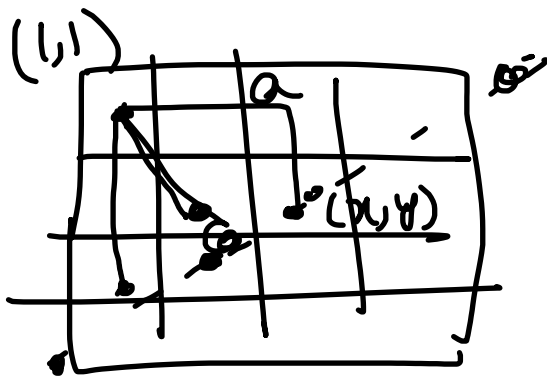
$$(x_{d_1}, y_{d_1}) \rightsquigarrow (x_{d_2}, y_{d_2})$$

$$(x_1, y_1)$$

$$(x_2, y_2)$$

↓
0

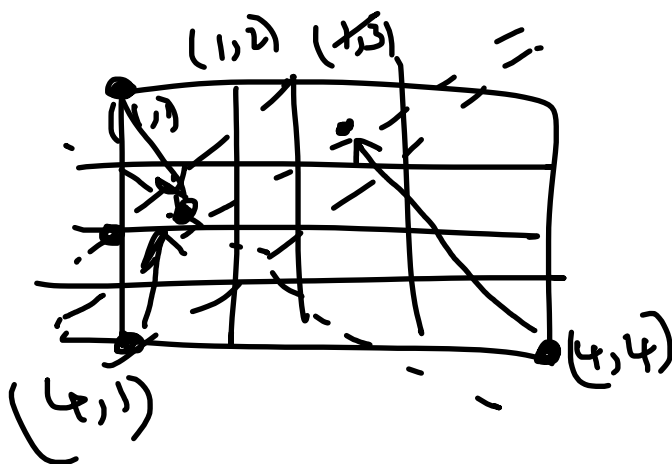
x 0



$$(x-1) + (y-1) = a$$

$$x + y - 2 = a$$

$$x + y = a + 2$$



Problem 3



Guessing The Greatest: <https://codeforces.com/contest/1486/problem/C2>

Main Idea



- Firstly query on the whole array to find the index of the global second maximum.
- Then perform another query to understand in which direction relative to the second maximum does the first maximum lie (left or right).
- Now binary search on the location of the first maximum.

$[\{a_1, a_2, a_3\}, \{a_4, \dots, a_n\}]$

$(1, n) \rightarrow ind, \quad \leftarrow \overset{ind}{=} \rightarrow$

$(1, ind) \rightarrow ind, //$

$(1, ind) \rightarrow \times ind //$
 $\quad \quad \quad = \underline{ind2}$
 right

$l = ind + 1$

$r = n, //$

$\underset{\uparrow}{mid} = (l + r) / 2, //$

(ind, mid)

$[a_1, a_2, \{a_3, \dots, \{a_{mid}, \dots, a_n\}\}]$
 $\quad \quad \uparrow \quad \quad \uparrow \quad \quad \uparrow \quad \quad \uparrow \quad \quad \uparrow$
 $\quad \quad ind2 \quad ind \quad \quad mid \quad \quad \underline{mid}, mid$

$a(\underline{ind}, \underline{mid}) \rightarrow \underline{ind}$

$ans = mid$
 $high = mid - 1, //$

$\boxed{T} \rightarrow a(ind, mid) \rightarrow ind, //$
 $\underline{F} \rightarrow a(\underline{ind}, \underline{mid}) \neq ind, //$
 $low = mid + 1$

$$l = 1$$

$$r = \text{ind} - 1$$



$$T \rightarrow a(\text{mid}, \text{ind}) \rightarrow \text{ind}$$

$$F \rightarrow a(\text{mid}, \text{ind}) \neq \text{ind}$$

$$\begin{aligned} &\rightarrow \text{ans} = \text{mid} \\ &\text{low} = \text{mid} + 1 \end{aligned}$$

$$\text{high} = \text{mid} - 1$$

